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# Cybersickness and User Acceptance in Virtual Reality Training: An analysis of Influencing Factors

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**Abstract**

Virtual Reality offers promising benefits for training, but its effectiveness is challenged by varying physical and cognitive demands, affecting users differently. These variations can lead to issues like cybersickness, impacting the overall success of VR-based training in modern smart factories. This paper investigates these challenges through a meta-analysis of studies assessing cybersickness using the Simulator Sickness Questionnaire (SSQ), across different domains and tasks, supported by the results of an ad-hoc VR experimental campaign with both adult and younger participants. Additionally, the Unified Theory of Acceptance and Use of Technology (UTAUT) is applied to explore barriers to adopting VR for training in industrial settings. Results show how cybersickness depends on several, interconnected factors, which vary based on user and task in analysis. Nonetheless, the study shows the benefit of using VR for training, and how performance can affect positively the willingness to use VR as a new training paradigm.

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*Keywords:* Virtual reality; Cybersickness; SSQ; UTAUT; Industry 5.0

## 1. Introduction

Virtual Reality (VR) is transforming workforce training by offering highly realistic, immersive, and interactive learning environments, enabling users to develop and reinforce their skills in safe, true-to-life scenarios while minimizing real-world risks [1]. Research has demonstrated VR's effectiveness in reducing training time, preventing operational errors, and enhancing overall product quality [2]. The technology has already proven valuable across diverse sectors, including healthcare [3], aviation [4], emergency response [5], and education [6]. However, the effectiveness of VR training experiences varies significantly based on their physical and cognitive demands. These demands are modulated by two key categories of factors: user-specific characteristics [7] such as age, cognitive capacity, physical

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condition, and motivation; and task-related variables [8] including session duration, required physical movements, and hardware specifications. Understanding the interplay between these factors is crucial for optimizing training outcomes. By carefully considering these relationships, trainers can customize session parameters to individual user needs, thereby minimizing adverse effects and maximizing learning effectiveness [9].

This paper provides a comprehensive analysis of user- and task-related factors influencing the acceptance of VR technology for training in industrial settings. Specifically, it presents a meta-analysis of cybersickness studies alongside findings from an on-site VR experiment conducted with both adult and younger participants. Furthermore, by applying the UTAUT framework to operators in the manufacturing sector, this study offers new insights into the factors shaping the willingness to adopt VR-based training.

The paper is organized as follows: Section 2 highlights the theoretical background; Section 3 streamlines the methodology followed to get the results, as shown in Section 4. Finally, discussions and conclusions are provided in Section 5, as well as the study's limitations and future research directions.

## 2. Theoretical Background

User acceptance is a critical determinant in the successful implementation of VR training solutions [10]. In the context of VR training, user acceptance encompasses both individual employees' and organizations' willingness to integrate VR technologies into their existing training frameworks. However, several technological challenges, particularly those related to user experience and side effects, can significantly impact this acceptance [8].

One of the most significant barriers to VR adoption is cybersickness, an adverse physiological response to immersive interfaces characterized by symptoms including fatigue, migraines, headaches, nausea, postural instability, and disorientation [11]. This phenomenon primarily stems from sensory conflicts between the vestibular system (responsible for balance, motion perception, and spatial orientation) and the visual system processing environmental cues through the central nervous system [12]. Research indicates that approximately 80% of users experience cybersickness symptoms within their first 10 minutes of VR exposure [12].

The evaluation of VR-related side effects employs both subjective and objective methodologies. Subjective assessments typically utilize standardized questionnaires such as SSQ [13], VRSQ [14], and CSQ [15]. Available studies in the literature, however, frequently focus on younger participants, limiting the generalizability of findings to the broader industrial workforce population [16].

The UTAUT model [17] has emerged as a prominent framework for evaluating users' willingness to adopt VR training. While this model has been extensively applied in medical, psychological, and educational contexts, its application in industrial training remains limited. A systematic literature search using the query (TITLE-ABS-KEY("virtual reality") AND TITLE-ABS-KEY(acceptability) OR TITLE-ABS-KEY(acceptance) OR TITLE-ABS-KEY(utaut) AND TITLE-ABS-KEY(manufacturing) AND TITLE-ABS-KEY(training) OR TITLE-ABS-KEY("workforce training") OR TITLE-ABS-KEY("operator* training")) yielded only 19 results, of which merely 8 were relevant upon screening. This scarcity of research highlights a significant gap in understanding end-user acceptance of VR technology in manufacturing workforce training, which is a factor that could substantially impact the technology's practical effectiveness.

## 3. Methodology

The research followed a three-phase approach to investigate VR training implementation in industrial settings: first, a systematic meta-analysis of cybersickness effects associated with head-mounted displays; second, on-site experimental tests with participants of different age groups; and third, a technology acceptance evaluation using the UTAUT model with potential end users from industrial organizations to assess VR training acceptability in manufacturing environments.

### 3.1. Meta-analysis

A systematic meta-analysis following PRISMA guidelines was conducted on VR and cybersickness studies. The steps include an identification phase, followed by screening and eligibility of the results, and paper inclusion. The flow diagram illustrated in Figure 1 shows the steps of the analysis.

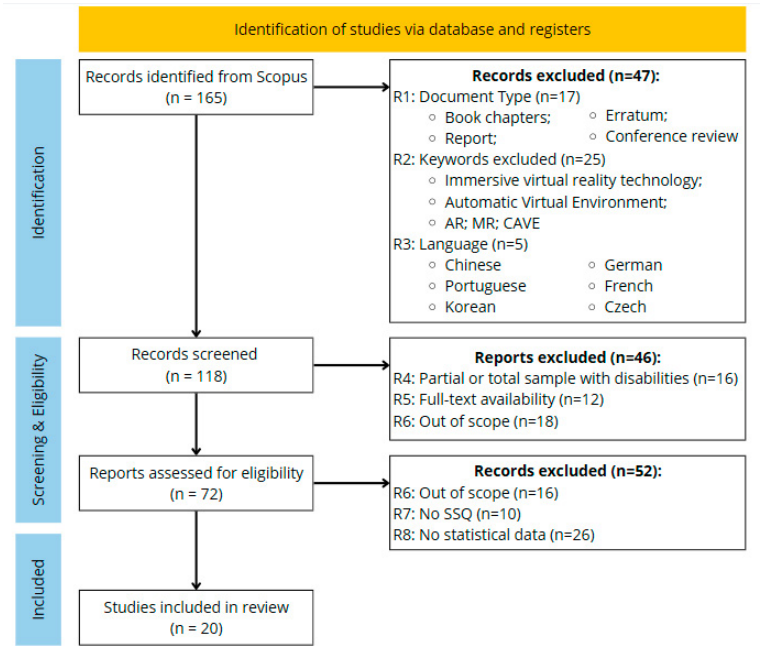


Fig. 1. PRISMA methodology

#### 3.1.1. Identification

The research was carried out on Scopus on the 21st of May 2024, using the following search query:

**Query**: TITLE-ABS-KEY ( cybersickness ) OR TITLE-ABS-KEY ( ”simulator sickness” ) OR TITLE-ABS-KEY ( ”virtual reality sickness” ) OR TITLE-ABS-KEY ( ”virtual reality induced sickness” ) OR TITLE-ABS-KEY ( ”motion sickness” ) AND TITLE-ABS-KEY ( hmd* ) OR TITLE-ABS-KEY ( ”head-mounted display” ) OR TITLE-ABS-KEY ( ”VR headset” ) AND TITLE-ABS-KEY ( ”virtual reality” ) OR TITLE-ABS-KEY ( ”virtual environment” ) AND TITLE-ABS-KEY ( training* ) OR TITLE-ABS-KEY ( learning )

This first research identified 165 contributions, which were filtered based on R1, R2, and R3 criteria described in Figure 1. The records excluded were 47, thus leaving 118 papers to be screened.

#### 3.1.2. Screening and eligibility

This phase involves two steps: screening titles and abstracts for relevance and assessing full-text eligibility of remaining papers.

**Screening** The 118 scientific contributions were analyzed and filtered using the R4, R5, and R6 criteria shown in Figure 1. Papers that satisfied R4 were 16, R5 were 12, while R6 in this phase were 18.

**Eligibility** A total of 72 papers were then assessed for eligibility and filtered based on R6, R7, and R8 criteria. An additional 16 papers were considered out of scope, while 10 did not use the SSQ questionnaire. This criterion was included to have the same sample valuation methodology among studies, providing mean data for the sub-scales or

the total score. Finally, among the paper that used the SSQ, 26 papers were excluded because of missing statistical measures.

#### 3.1.3. Included papers

After applying filters and exclusion criteria to the initial 165 papers, 20 were found relevant to this research. These included research published in journals, conference papers, and reviews, as well as papers that deal with rehabilitation with tests carried out on healthy samples.

#### 3.1.4. Data collection

The meta-analysis synthesized data across multiple dimensions, including application field, VR purpose, and SSQ statistical values. Studies were included in the analysis even when they reported only total cybersickness scores without symptom-specific breakdowns. For studies providing detailed symptom cluster values, the total score was calculated using the standardized formula: TOT = $(N/9.54+O/7.58+D/13.92)*3.74$, where N, O, and D represent nausea, oculomotor, and disorientation clusters respectively. For studies reporting multiple tests or comprising collections of research, data were consolidated by computing overall mean values. To facilitate meaningful comparisons, studies reporting session duration data were categorized into four-time intervals: $\leq 10$; $10 < t \leq 25$; $25 < t \leq 60$; $> 60$.

### 3.2. On-site test for cybersickness assessment

The experimentation was conducted at the University of Calabria with voluntary participants. Data collection centered on SSQ assessments, which is considered the standard for cybersickness evaluation [18], along with participant demographics (age, sex), physical characteristics (vision status, eyewear use), and session parameters (duration, time of day, prior VR experience).

#### 3.2.1. The sample

The study sample, detailed in Table 1, consisted of voluntary participants from two age groups: adults (40-65 years) employed across various sectors (renewable energy, medical, educational) and young adults (18-30 years) enrolled in engineering programs at Italian universities.

Table 1. On-site experimentation sample data. Experience: recurrent [>5]; sporadic [$\geqslant$ 1]

| Sample | Number | Age | Sex [%] | Reported sight issues [%] | Experience with VR [%] |
|---|---|---|---|---|---|
| Adult | 7 | Mean: 54,00<br>Stand. Dev.: 10,73 | Male: 71,43<br>Female: 28,57 | Yes: 28,57<br>No: 42,55 | Recurrent: None<br>Sporadic: 42,86<br>None: 57,14 |
| Young | 47 | Mean: 22,30<br>Stand. Dev.: 2,67 | Male: 68,09<br>Female: 31,91 | Yes: 57,45<br>No: 42,55 | Recurrent: 4,26<br>Sporadic: 27,66<br>None: 68,09 |

#### 3.2.2. The virtual reality session

The experiment utilized Meta Quest 2 with Job Simulator, a game simulating workplace activities through tasks analogous to real industrial operations (dragging, bending, rotating). Participants, positioned stationary in a room, relied solely on in-game assistance to complete self-paced tasks. As shown in Figure 2, users interacted with the virtual environment, with sessions averaging 23.74 minutes and ending at the participant's discretion based on symptom onset.

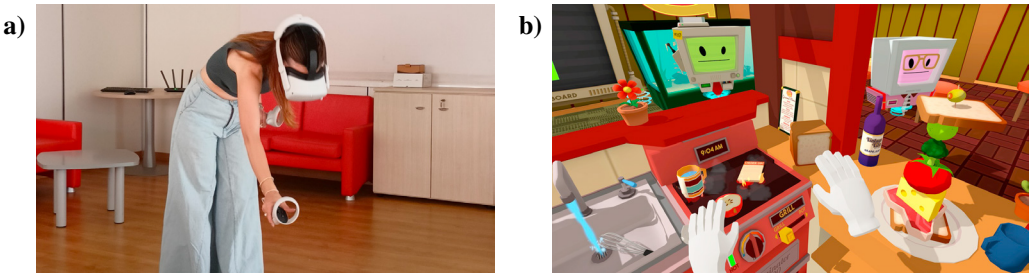


Fig. 2. On-site VR session set-up. a) Participants executing a task; b) Game-play display

### *3.2.3. Simulator Sickness Questionnaire*

To assess cybersickness levels in the sample group, the SSQ developed by Kennedy and Lane [19] was administered immediately following VR exposure. The SSQ evaluates sixteen standardized symptoms (including general discomfort, fatigue, headache, and others) using a 4-point severity scale. These raw scores are then weighted and aggregated into three primary symptom clusters (N,O,D). The questionnaire administration was deliberately scheduled five minutes post-session to allow for the full manifestation of potential symptoms. The relationship between recorded factors and cybersickness severity was then analyzed using a Regression Model to quantify their respective impacts on total SSQ scores.

## *3.3. Technology acceptance evaluation*

The UTAUT questionnaire was administered to potential end users through both LinkedIn professional network and direct interviews. The survey was conducted in Italian to accommodate the target population, which comprised 40 Italian manufacturing professionals spanning various roles within the industry, including specialized operators, industrial technicians, mechanical technicians, foremen, metal-mechanic workers, and maintenance personnel. The study aimed to assess the acceptance of VR-based training compared to conventional training methods, encompassing classroom instruction, on-the-job training, and mentorship programs. Prior to administering the UTAUT questionnaire, demographic and professional data were collected from participants, including age, field of work, contractual level, and previous exposure to VR technology.

### *3.3.1. Hypotheses and items*

The research framework comprised twenty-one items, structured to validate the hypothesized relationships between key predictors and user acceptance. The study examined four primary predictors theorized to influence users' behavioral intention (BI): performance expectancy (PE), effort expectancy (EE), social influence (SI), and facilitating conditions (FC). Additionally, participant age was incorporated as a moderating variable in the analysis framework, as illustrated in Figure 3. The specific hypotheses associated with each predictor are detailed in Table 2. Participants evaluated each of the twenty-one items using a 6-point Likert scale, enabling quantitative assessment of the proposed relationships between variables.

Table 2. UTAUT hypotheses for predictors

| Predictors | ID | Hypotheses |
|---|---|---|
| PE | H1 | PE has a positive effect on the behavioral intention of workers to train with VR tools |
| EE | H2 | EE has a significant effect on the BI of workers to train with VR tools |
| SI | H3 | SI has a significant effect on the BI of workers to train with VR tools |
| FC | H4 | FC have a significant effect on the BI of workers to train with VR tools |

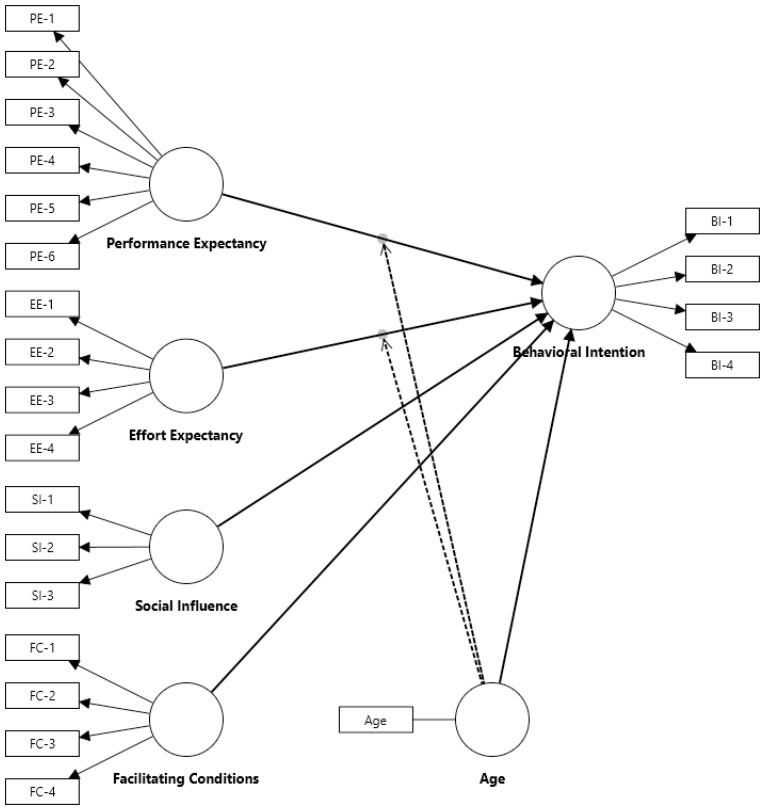


Fig. 3. UTAUT framework for PLS-SEM analysis

### *3.3.2. Data management and measurement model results*

The analysis of questionnaire data began with a preliminary screening phase to identify and remove potential outliers. This process led to the exclusion of three responses where participants had assigned uniform ratings across all items, indicating potentially non-discriminating responses. The data analysis followed the methodological guidelines established by Hair et al. [20], employing Partial Least Squares-Structural Equation Modeling (PLS-SEM). Following these validation procedures, the analysis determined the significance and relevance of relationships between each predictor and behavioral intention (BI).

## 4. Results

### *4.1. Meta analysis for identifying task-related factors affecting cybersickness*

The meta-analysis partial extracted data are presented in Table 3. Medical and technology applications dominate the literature, while manufacturing scenarios are notably absent. Average symptom scores (N=23.60, O=22.38, D=34.36, TOT=23.73) suggest that disorientation is typically the most pronounced effect.

The relationship between VR exposure duration and cybersickness symptoms is proved to be complex and non-linear. While sessions exceeding 60 minutes did not generate the most severe symptoms, sessions shorter than 10 minutes produced unexpectedly high symptom scores [12]. Sessions ranging from 10 to 60 minutes typically elicited moderate symptom levels, suggesting potential physiological adaptation to virtual environments. This intricate temporal distribution highlights the multifaceted nature of cybersickness, indicating that different physiological responses may peak at varying intervals throughout VR exposure.

The meta-analysis revealed distinct patterns in cybersickness symptoms across device generations. Earlier-generation devices (HTC Vive, Oculus Rift CV1, Oculus Rift DK2, and ExpeditionDI) consistently produced higher symptom scores, with disorientation being particularly pronounced. In contrast, contemporary devices (HTC Vive Pro Eye, Meta Quest 2, HTC Vive Cosmos, DJI FPV Goggles V2, and Pimax5k+) demonstrated improved user tolerance, maintaining lower and more consistent symptom levels even during extended VR exposure.

Table 3. Meta-analysis of cybersickness assessment in the literature.

| Author | Field | Duration | N [mean] | O [mean] | D [mean] | TOT [mean] |
|---|---|---|---|---|---|---|
| Ritter et al. [13] | Sport | | 15,71 | 16,05 | 23,33 | 20,35 |
| Zorzi et al. [3] | Medical | | 74,2 | 53,9 | 106,7 | 84,4 |
| Ito et al. [21] | Sensitization | | 5,55 | 13 | 13,05 | 8,1 |
| Morizio et al. [22] | Medical | 20 | | | | 2,825 |
| Teixeira et al. [23] | Technology | 14 | 48,65 | 37,142 | 68,21 | 55,72 |
| Kaufeld et al. [24] | Technology | 15 | 22,43 | 17,23 | 25,13 | 24,04 |
| Solmaz et al. [25] | Engineering | 20 | | | | 17,53 |
| Halkola et al. [26] | Technology | 4 | | | | 36,23 |
| Sivenas et al. [27] | Pedagogy | 15 | | | | 20,13 |
| Szpak et al. [28] | Technology | 50 | 20,074 | 24,003 | 29,29 | 20,03 |
| Auer et al. [4] | Aviation | 75 | 10,41 | 21,36 | 24,04 | 23,81 |
| Grassini et al. [29] | Technology | 10 | | | | 20,57 |
| Saredakis et al. [30] | Technology | | 16,723 | 17,092 | 23,499 | 28 |
| Sakhare et al. [31] | Medical | 2,5 | 25 | 17 | 23 | 25 |
| Kern et al. [32] | Medical | 7,5 | 15,1 | 12,42 | 11,21 | 14,23 |
| Di Loreto et al. [33] | Operation | 15 | | | | 1,39 |
| Maloca et al. [34] | Medical | 5 | | | | 5,8 |
| Kim et al. [35] | Medical | 25 | | | | 12,2 |
| Papachristos et al. [36] | Technology | 6 | 15,9 | 28,805 | 49,65 | 33,785 |
| Taylor & Barnett [37] | Military | | 13,42 | 22,5 | 15,23 | 20,45 |
| **Results** | | | **23,60** | **23,38** | **34,36** | **23,73** |

### *4.2. User-related factors affecting cybersickness and technology acceptance*

The results of the SSQ and the UTAUT carried out will be described in the following sections, highlighting the influencing factors on cybersickness and technology acceptance.

#### *4.2.1. On-site test for determining factors impacting cybersickness*

The regression model revealed several significant correlations between user characteristics and cybersickness severity. Age emerged as a complex predictor, showing significant variations within both young and adult groups (p-value = 0.019). A notable quadratic effect (p-value = 0.012) indicates that the relationship between age and cybersickness is non-linear, with symptom peaks observed at specific ages within both younger and older groups. While the overall age group classification (adult and young) showed marginally significant effects (p-value = 0.075), the interaction between age and time of day proved highly significant (p-value = 0.002), suggesting that circadian rhythms and fatigue patterns may influence cybersickness susceptibility differently across age groups.

User experience demonstrated a significant non-linear effect (p-value = 0.006), indicating that while initial VR exposure might increase cybersickness susceptibility, the extended experience could lead to adaptation, potentially reaching an optimal point of minimal symptoms. The combination of experience and duration of VR sessions only marginally affect (p-value = 0.064) cybersickness effect.

Conversely, the interaction between experience and vision conditions (p-value = 0.029) suggests that visual characteristics significantly modulate the benefits of VR experience. The data shows that users with recurrent VR experience and vision issues show notably higher disorientation scores (D = 55.68), compared to their oculomotor disturbances (O = 11.37), while maintaining moderate nausea levels (N = 23.85). Users with sporadic experience demonstrate the highest overall symptom scores, particularly those with vision issues (TOT = 50.49). Users with no prior VR experience show relatively moderate and consistent symptom levels across all measures, regardless of vision status.

Furthermore, the three-way interaction between session duration, experience, and vision status proved significant (p-value = 0.032), highlighting the complex interplay of these factors.

As illustrated in Figure 4, the on-site experimental results generally aligned with meta-analysis findings, though with notably higher total scores (29.92 versus 23.73) and disorientation values (37.61 versus 34.36).

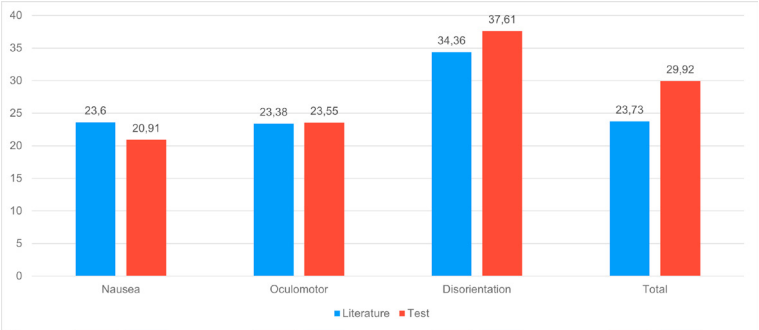


Fig. 4. Comparison form overall SSQ results from the meta-analysis and on-site tests

### 4.2.2. *UTAUT for determining factors impacting technology acceptance*

The UTAUT measurement model demonstrated overall positive results, though some values were borderline. Model reliability was enhanced by removing two facilitating conditions (FC) constructs based on preliminary analysis. The study surveyed industrial operators with a mean age of 34.10 years, with 28% reporting prior VR experience. Participants represented various professional levels according to Italian contract classifications: 20.5% held technical roles (C3), while specialist (B1) and management (B2) positions accounted for 20.5% and 15.4% respectively, with remaining participants distributed evenly across other classifications.

Analysis of the proposed hypotheses (Table 2) revealed varying levels of support, as detailed in Table 4. Facilitating conditions emerged as the strongest predictor of behavioral intention (BI) with a coefficient of 0.405, followed by performance expectancy (PE) at 0.379, both achieving statistical significance (p ¡ 0.05). FC-related factors, particularly retained knowledge and training session enjoyment, demonstrated substantial influence on VR adoption intentions. Similarly, PE factors, including perceived utility, productivity, effectiveness, and efficiency compared to traditional training methods, showed significant positive impact on users' behavioral intentions toward VR technology adoption.

Table 4. UTAUT predictor influence assessment

| | Original Sample (O) | Sample mean (M) | Standard deviation (STDEV) | T statistics (O/STDEV) | P-values | Result |
|---|---|---|---|---|---|---|
| PE– >BI (H1) | 0.379 | 0.395 | 0.159 | 2.383 | 0.017 | Supported |
| EE– >BI (H2) | 0.084 | 0.057 | 0.206 | 0.406 | 0.685 | Not Supported |
| SI– >BI (H3) | 0.095 | 0.079 | 0.129 | 0.731 | 0.465 | Not Supported |
| FC– >BI (H4) | 0.405 | 0.421 | 0.188 | 2.149 | 0.032 | Supported |

## 5. Discussions and conclusions

This study advances our understanding of VR implementation challenges by examining cybersickness factors and user acceptance in manufacturing contexts, providing guidelines for effective training program design.

To optimize training effectiveness and minimize side effects, sessions should be configured considering multiple factors. For adult users, morning sessions are recommended given the significant effect of exposure time on cybersickness symptoms (p-value = 0.002). Session duration should be modulated based on user experience (p-value = 0.006) and vision conditions (p-value = 0.029), with initial sessions kept shorter to allow for adaptation.

To enhance user acceptance, organizations should focus on two key aspects: providing robust technical support and clearly demonstrating training benefits, as indicated by the significant influence of facilitating conditions (p-value = 0.405) and performance expectancy (p-value = 0.379) in the UTAUT model. When available, newer generation VR devices should be preferred as they consistently show lower cybersickness scores across extended sessions, though regular breaks and user monitoring remain important regardless of equipment specifications.

However, this research presents some limitations. The regression model used to analyze the on-site results (R-sq = 86.02%, p-value = 0.056) suggests the need for further validation with larger samples. While the model explains a substantial portion of cybersickness variance, its borderline significance indicates that additional factors might influence user responses. The small sample of experienced VR users constrains our understanding of expertise's role in mitigating cybersickness, while the focus on Italian participants limits the generalizability of acceptance findings. Future research should explore these aspects with larger, more diverse samples across different cultural contexts and experience levels.

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